The Monotone Follower Problem

in Stochastic Decision Theory

Singular Stochastic Control & Variational Inequalities

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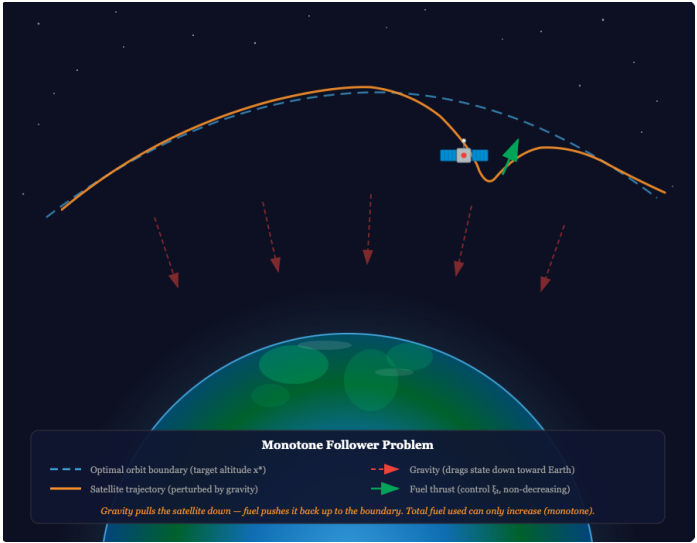
Outline

1. Introduction & Motivation
2. Summary & Open Problems

Singular control problem: a foundational example.

- Suppose you have a space satellite that orbits the Earth, but its trajectory is perturbed by random forces.
- You can use fuel to adjust the satellite's position, but since fuel is costly/limited, you want to use it as efficiently as possible.
- You want to keep the satellite within its orbit by applying a control (fuel usage) that pushes it back to the optimal orbit, while minimizing the total expected cost of fuel and deviation from the target altitude.
- This problem of interested in the first formal problem of **singular stochastic control**, [1], [2] (J.A. Bather, H. Chernoff, Sequential decisions in the control of a spaceship (with finite fuel), 1967).

The satelite control problem.



Stefan Problem: Ice Melting

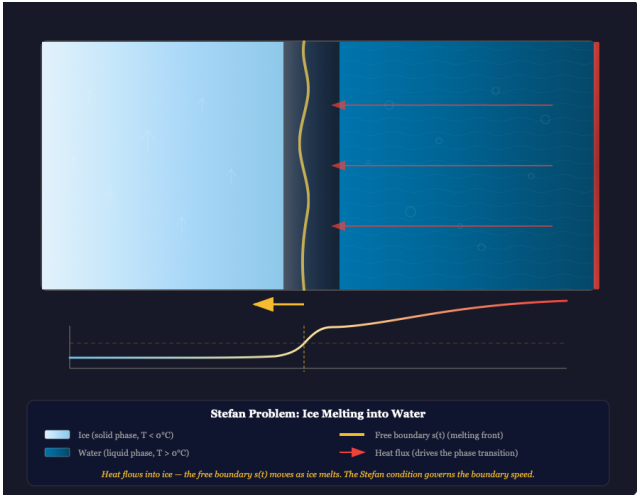


Figure: The free boundary $s(t)$ advances as heat melts ice into water. [Click for animation]

What Is the Monotone Follower Problem?

- A **stochastic control** problem where the controller chooses a *non-decreasing*, adapted process $\xi = \{\xi_t\}_{t \geq 0}$.
- The state evolves as

$$X_t = x + \int_0^t b(X_s)\,\mathrm{d}s + \int_0^t \sigma(X_s)\,\mathrm{d}W_s + \xi_t,$$

  where W is a standard Brownian motion.
- The controller seeks to **minimise** a cost functional that balances a running cost against the cumulative cost of intervention:

$$J(x;\xi) = \inf_{\xi} \mathbb{E}_x\left[\int_0^T f(X_t)\,\mathrm{d}t + \hat{\xi}_T\right].$$

- where $\hat{\xi}_T = \xi_T^+ + \xi_T^-$ is the net control applied up to time T.

Historical Context

- **Bather & Chernoff (1967a, 1967b):** Early formulation as a sequential decision problem (finite-fuel spaceship control).
- **Beneš, Shepp & Witsenhausen (1980):** Explicit solutions for several singular control problems.
- **Karatzas (1981):** Monotone follower formulation and reduction to a free-boundary problem.
- **Karatzas (1983):** A class of singular stochastic control problems.
- **Karatzas & Shreve (1984):** Connection between optimal stopping and singular stochastic control.
- **El Karoui & Karatzas (1988):** General existence results via convex duality.
- Modern extensions: multi-dimensional controls, regime switching, mean-field interactions, Lévy-driven dynamics.

Motivating Applications

Economics & Finance

- Irreversible investment under uncertainty
- Optimal dividend policy
- Inventory management
- Real options with sunk costs

Engineering & Operations

- Spacecraft fuel-optimal attitude control
- Dam and reservoir management
- Queueing system regulation
- Tracking a stochastic signal

A solvable example

- We first study the work in [3].
- Authors want to solve the following problem:

$$\inf_{\xi \in \Xi} \mathbb{E}\left[\int_0^T (x + W_t - \xi_t)^2 \, dt\right]$$

where the class of controls Ξ is given by

$$\Xi = \left\{ \xi : \begin{array}{l} \xi \text{ is adapted to } (x, W), \\ \xi_0 = 0, \\ \theta_0 \leq \xi_t \leq \theta_1 \end{array} \right\}.$$

- A closed-form solution is obtained $\theta_0 = 0$, $\theta_1 = \infty$.

- The Bellman (HJB) equation which solution v is defined as

$$v(\tau, x) = \inf_{\xi \in \Xi} \mathbb{E}\left[\int_{T-\tau}^{T} (x_t)^2 \mathrm{d}t \middle| x_{T-\tau} 0 = x \right].$$

is given by

$$v_t = x^2 + \frac{1}{2} v_{xx} + \inf_{0 \leq u \leq \infty} (-u v_x)$$

See that $v_x \leq 0$

$$\begin{cases} v_x < 0 \implies u^* = 0, & \text{not exercising zone} \\ v_x = 0 \implies u^* \in [0, \infty), & \text{exercise zone} \end{cases}$$

- The problem is then to vind the couple of functions $(v, f(\tau))$ that solve the PDE

$$\begin{cases} v_t = x^2 + \frac{1}{2} v_{xx}, & x < f(\tau) \\ v_x = 0, & x \geq f(\tau) \end{cases}$$

Numerical Methods

Method	**Approach**	**Key Reference**
Finite differences	Discretise HJB / VI	Kushner & Dupuis (2001)
Penalty method	Approximate ξ by ε^{-1} terms	Lions & Sznitman (1984)
Policy iteration	Iterate on free boundary	Huang & Forsyth (2012)
Monte Carlo	Regression-based (Longstaff–Schwartz)	Belomestny et al. (2015)
Deep learning	Neural net for V or ξ^*	Han et al. (2018)

- Curse of dimensionality is severe for $d \geq 3$.
- Deep learning approaches scale better but lack convergence guarantees.

Summary

1. The **monotone follower** is a canonical singular stochastic control problem with wide applicability.
2. The value function satisfies a **variational inequality** with free-boundary structure.
3. A powerful **duality** links singular control to optimal stopping via differentiation of the value function.
4. Explicit solutions exist in the linear–quadratic Gaussian case (Skorokhod reflection at x^*).
5. Extensions to multi-dimensional, Lévy, and mean-field settings remain active research areas.

Open Problems

- **Regularity of the free boundary** in multi-dimensional singular control.
- **Non-convex running costs:** breakdown of the classical smooth-fit approach.
- **Model uncertainty / robustness:** minimax monotone follower under ambiguity.
- **Mean-field games** with singular controls: existence and uniqueness of Nash equilibria.
- **Data-driven approaches:** reinforcement learning for singular control without model knowledge.

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Thank You!

Questions & Discussion

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